

SecureAuth - JWT Based User Authentication and Authorization System

Project Overview

SecureAuth is a secure user management system built using Spring Boot. The system allows users to register, log in, verify email addresses, reset passwords, and manage profiles. Authentication is handled using JWT tokens and authorization is managed through role-based access control.

Tech Stack

Frontend: React or HTML/CSS/JavaScript

Backend: Spring Boot

Database: PostgreSQL

Security: Spring Security + JWT

Email Service: Spring Mail (SMTP)

Build Tool: Maven

Version Control: Git and GitHub

Features

1. User Registration – Users create accounts with email and password.
2. User Login – Registered users authenticate using credentials.
3. JWT Authentication – Secure token-based authentication for APIs.
4. Role-Based Access Control – Different permissions for Admin and User roles.
5. Welcome Email – Automatically sends an email after successful registration.
6. Login Alert Email – Sends an email whenever a login occurs.
7. Email Verification – Users verify ownership of their email address before accessing all features.
8. Forgot Password with OTP – OTP is sent to the registered email for password reset.
9. Profile Management – Users can update profile details.
10. Change Password – Users can securely change their passwords.

Database Tables

Users Table, Roles Table, Email Verification Tokens Table, Password Reset OTP Table.

System Flow

User → Register → Email Verification → Login → JWT Token Generated → Access Protected APIs → Manage Profile.

Forgot Password Flow: User → Request OTP → Receive Email → Verify OTP → Set New Password.

Flowchart

Start → Register → Verify Email → Login → Generate JWT → Access Application → Logout → End.

Admin Flow: Login → JWT Validation → Admin Dashboard → Manage Users.

Future Enhancements

Google Login, Two-Factor Authentication (2FA), Refresh Tokens, Audit Logs, Docker Deployment, Cloud Hosting.